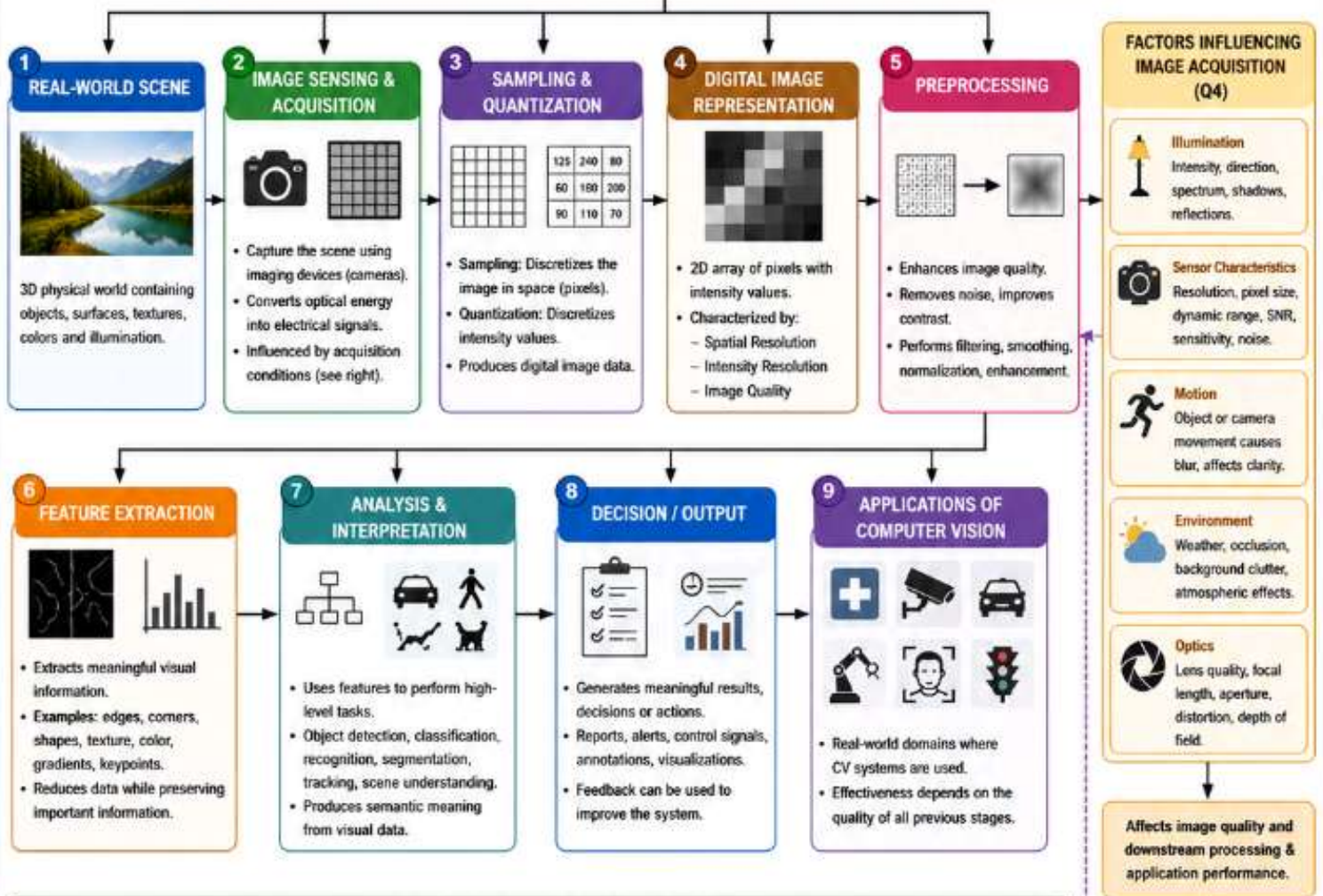


# COMPUTER VISION – CONCEPT MAP (CO1)

**Goal:** To enable computers to acquire, process, analyze and understand visual data from the real world and produce meaningful information for useful applications.



## KEY RELATIONSHIPS (Q1 & Q2 & Q3)



## IMAGE QUALITY DETERMINANTS (Q3)



## COMPUTER VISION CAPABILITIES (Q5)

- Image Acquisition
- Preprocessing
- Feature Extraction
- Pattern Recognition / Classification
- Segmentation & Tracking
- 3D Reconstruction
- Learning & Adaptation
- Real-time Processing

## INFLUENCE APPLICATION SELECTION (Q5)

Capabilities determine which applications can be solved effectively and the level of accuracy, speed, robustness and reliability that can be achieved.

## RECENT LITERATURE SUPPORTING THE CONCEPTS (At least Five Sources)

- [1] Jia, X., Shelhamer, E., Donahue, J., Karayev, S., Long, J., Girshick, R., Guadarrama, S., & Darrell, T. (2014). Caffe: Convolutional Architecture for Fast Feature Embedding. *ACM MM*, 675–678. (Deep features and representation)
- [2] Krizhevsky, A., Sutskever, I., & Hinton, G. E. (2017). ImageNet Classification with Deep Convolutional Neural Networks. *Communications of the ACM*. (Impact of deep learning in CV)
- [3] Gonzalez, R. C., & Woods, R. E. (2018). *Digital Image Processing* (4th Ed.). Pearson. (Foundations of image acquisition, sampling, quantization, resolution, and quality)
- [4] Szeliski, R. (2022). *Computer Vision: Algorithms and Applications* (2nd Ed.). Springer. (Pipeline, features, and visual interpretation)
- [5] Litjens, G., Kooi, T., Bejnordi, B. E., Setio, A. A. A., Ciampi, F., Ghafoorian, M., van der Laak, J. A., van Ginneken, B., Sánchez, C. I. (2017). A survey on deep learning in medical image analysis. *Medical Image Analysis*, 42, 60–88. (Applications in medical imaging)

**LEGEND:** → Flow / Process    - - - - - Influence / Effect    — Relationship    Major Concept / Stage